

Problem 1, Start to Finish

Maya's investment plan, worked through the seven-step method. IE 5311-001 (D01) Principles of Optimization, Fall 2026.

Companion to the method sheet. Every step below is one of the seven moves, applied to a real problem, with the reasoning left in rather than cleaned away.

The story, in plain words

Maya has \$5,000 and five years. At the start of each year she may place money into a deposit that locks for 1, 2 or 3 years. When a deposit matures the money comes back and she may reinvest it. She wants the most cash possible at the end of Year 5.

Instrument	Term	Total return over the term	Available from
Frontier 1-year	1 yr	1.04	start of Year 1
Frontier 2-year	2 yr	1.09	start of Year 1
Pioneer 3-year	3 yr	1.15	start of Year 2

That last cell is the whole problem hiding in plain sight. Hold the thought.

Step 0. Fix the time convention first

This step is not in the generic checklist, because it only appears in problems that move through time. When it does appear, do it before anything else, because every later decision depends on it.

The question: does "period 3" mean the *start* of Year 3 or the *end* of Year 3?

We chose: **epoch t is the beginning of Year t** . So t runs 1 to 5, and we add $t = 6$ to mean the end of Year 5, which is the moment the answer is measured. Everything then follows mechanically:

An instrument of term L started at epoch t matures at epoch $t + L$, and is admissible only if $t + L \leq 6$.

That one line does all the filtering. A 3-year deposit started at epoch 4 would mature at epoch 7, and there is no epoch 7, so Maya's money would still be locked when the horizon arrives. Not admissible. You never have to reason about it case by case; the inequality handles it.

Get this convention wrong and every number after it is wrong, with nothing in the solver to warn you. Write it down explicitly in your answer. Two sentences is enough.

Step 1. Find the decision

Apply the test: if I handed you these numbers, could you walk into the bank and do it?

- "Put \$5,000 into the two-year deposit at the start of Year 1" — yes — **decision**
- "The two-year deposit returns 9%" — you cannot negotiate it — **parameter**
- "End Year 5 with \$6,267.50" — that is the consequence — **result, not a variable**

So the decision is: how many dollars go into which instrument, at which epoch.

The second decision most people miss

What if Maya simply holds cash, and does nothing with some of it for a year? You could argue she never would, because every instrument pays a positive return. The argument is correct. Do not bake it into the model anyway. Put in a variable for it:

w_t = dollars held idle from epoch t to epoch $t+1$

Then when the solver returns $w_t = 0$ at every epoch, "she reinvests everything" is a **result you proved** rather than an assumption you smuggled in. That distinction is worth marks and costs one extra variable.

Step 2. Index it

What does the decision vary over? Two things: which epoch, and which instrument. Two things means two subscripts.

$x_{\{t,k\}}$ = dollars placed at epoch t into an instrument of term k

The idle-cash decision varies over one thing, the epoch, so it takes one subscript: w_t .

Notice what we did not write: $x_1, x_2, \dots, x_{11}$. There are eleven admissible combinations here. Naming them one by one would work, would be unreadable, and would break the moment a fourth instrument appeared.

Step 3. The tables, before any math

Sets

Symbol	Definition
$T = \{1, \dots, 5\}$	decision epochs, the beginning of each year
$K = \{1, 2, 3\}$	instrument types, indexed by term in years
A	the admissible pairs (t, k) with $t \geq a_k$ and $t + L_k \leq 6$

The set A is doing real work. It is where the time convention gets enforced, once, instead of being scattered through the constraints.

Parameters

Symbol	Definition	Value
L_k	term of instrument k , years	1, 2, 3
r_k	total return multiplier over the full term	1.04, 1.09, 1.15
a_k	first epoch instrument k is available	1, 1, 2
B	initial endowment, dollars	5,000
H	horizon epoch	6

$a_3 = 2$ is how "Pioneer opens at the start of Year 2" becomes mathematics. One number in a table.

Decision variables

$x_{\{t,k\}} \geq 0$ dollars, for all (t, k) in A
 $w_t \geq 0$ dollars, for all t in T

Now count what A actually contains

A two-minute hand check that catches convention errors.

k	condition	admissible t	count
1 (1-yr)	$t + 1 \leq 6$	1, 2, 3, 4, 5	5
2 (2-yr)	$t + 2 \leq 6$	1, 2, 3, 4	4
3 (3-yr)	$t \geq 2$ and $t + 3 \leq 6$	2, 3 only	2

Eleven pairs. And look at the last row. The Pioneer certificate has **exactly two possible start dates in the entire problem**. It cannot start at epoch 1, because it is not open yet, and it cannot start at epoch 4 or later, because it would not mature in time. So before solving anything you already know the interesting question is roughly "epoch 2, epoch 3, or not at all".

Step 4. The objective

What is the final cash? Everything that matures *exactly* at the horizon, plus any cash sitting idle at the end.

$$\max \sum \text{over } (t, k) \text{ in } A \text{ with } t + L_k = H \text{ of } r_k x_{\{t, k\}} + w_5$$

The condition $t + L_k = H$ is the whole trick. Only deposits maturing precisely at epoch 6 pay into the final total. A deposit maturing at epoch 4 does not appear in the objective at all. It appears in the balance constraint at epoch 4, gets reinvested, and reaches the horizon through whatever it was reinvested into.

That is the part that confuses people. **Money reaches the objective only through the last hop**. Everything earlier is bookkeeping.

Step 5. The constraints

There is only one idea here, said five times. In words: at every epoch, the money you put out equals the money you have available.

At epoch 1 the money available is just the endowment.

$$\sum_k x_{\{1, k\}} + w_1 = B$$

At epochs 2 through 5 it is whatever matured just now plus whatever you carried forward.

$$\sum_k x_{\{t, k\}} + w_t = \sum \text{over } (s, k) \text{ with } s + L_k = t \text{ of } r_k x_{\{s, k\}} + w_{\{t-1\}}$$

Read it left to right: what I allocate now, plus what I hold back, equals what just matured, plus what I carried in.

This is **flow conservation on a timeline**. Structurally it is the same equation as the network flow constraint in Problems 2 and 3, out minus in equals b, where b is +\$5,000 at epoch 1 and zero everywhere after. Once you see that, three of the four homework problems are the same object.

Trace it with the actual answer, so it stops being abstract

Epoch	What matured	Carried in	Available	Allocated	Idle
1	—	—	\$5,000 endowment	\$5,000 into 2-yr	0
2	nothing	0	0	0	0
3	$5,000 \times 1.09 = \mathbf{\$5,450}$	0	\$5,450	\$5,450 into 3-yr	0
4	nothing	0	0	0	0
5	nothing	0	0	0	0
6	$5,450 \times 1.15 = \mathbf{\$6,267.50}$	—	<i>horizon</i>	—	—

Every row balances. Epoch 2 looks empty, and that is correct. The money is locked inside the two-year deposit, so there is genuinely nothing to decide that year.

Step 6. Domains

$$\begin{aligned} x_{\{t,k\}} &\geq 0 && \text{for all } (t,k) \text{ in } A \\ w_t &\geq 0 && \text{for all } t \text{ in } T \end{aligned}$$

Continuous, not integer. Money is divisible: you can deposit \$5,450.00 or \$3,217.63. **Nothing here needs to be a whole number**, so do not make it one. Forcing integrality would be a modelling error that also makes the problem harder to solve, for nothing.

Step 7. Classify it, on the slide 8 to 10 taxonomy

Question	Answer
Constrained?	Yes
Variable type and size	continuous, 11 x variables plus 5 w variables, 16 in total
Objective type	linear
Constraint type and size	linear, 5 equality rows plus nonnegativity
Parameters	deterministic , every rate known up front
Stages and players	single stage, one player

So it is a plain linear program, the smallest and friendliest object in this course. Compare with Problem 3, where the same six questions answer "stochastic, two-stage" and the whole solution method changes.

Sanity check before solving

Every admissible plan is a way of tiling the five years with blocks of length 1, 2 and 3, where no 3-block may start at epoch 1. So the value of a plan is the product of its multipliers, times \$5,000. There are only a handful, and you can check the answer by hand.

Plan, terms in order	Multiplier	Final
1,1,1,1,1	$1.04^5 = 1.2167$	\$6,083.26
1,1,1,2 in any order	$1.04^3 \times 1.09 = 1.2261$	\$6,130.51
1,2,2 in any order	$1.04 \times 1.09^2 = 1.2356$	\$6,178.12
1,1,3 or 1,3,1	$1.04^2 \times 1.15 = 1.2438$	\$6,219.20
2 then 3	$1.09 \times 1.15 = 1.2535$	\$6,267.50
3 then 2	—	not allowed , the 3-year cannot start at epoch 1

The winner is two-year then three-year. Notice the row underneath it. Three-then-two would give the same 1.2535 multiplier, but Pioneer is not open at epoch 1. The availability date is not decoration: it removes one of only two ways to reach the best multiplier, and it is exactly the kind of detail a problem statement buries in a subordinate clause.

Doing this table by hand before you solve is not wasted time. It is how you find out your model is wrong when it returns \$6,219.20 instead of \$6,267.50.

Reading the answer properly

The solver returns \$6,267.50, with this plan:

Epoch	Term	Amount	Matures at
1	2 years	\$5,000.00	epoch 3
3	3 years	\$5,450.00	epoch 6

No idle cash at any epoch, which confirms what we predicted but did not assume.

Now check the **shadow price** on the epoch-1 balance constraint. It comes back at **1.2535**, which is 1.09 times 1.15: the compounded multiplier of the winning plan, exactly. The economic reading is that one extra dollar handed to Maya at the start of Year 1 becomes \$1.2535 at the horizon, because she would route it through the same best chain.

When the dual independently reproduces a number you can derive by hand from the primal, you have genuine confirmation rather than a solver you are trusting on faith. That is what shadow prices are for, and it is the kind of thing an oral exam asks about.

What transfers to any problem like this

Anything with money, inventory or stock moving through time is this same problem in a costume. The pattern:

1. **Fix the time convention first**, and write it down. Start of period or end of period. Everything downstream depends on it.
2. **One balance equation per period**. What I use now equals what became available now plus what I carried in.
3. **Availability windows and term lengths live in the set definition**, not in the constraints.
4. **Only the final hop enters the objective**. Everything else flows through balance.
5. **Put in a "do nothing" variable** and let the solver prove it stays at zero.

Swap deposits for warehouse inventory, and the return multiplier for a spoilage rate. Same five moves, same model, a different table of numbers.